

ChronicGuard AI

Evaluating LLM-Assisted Patient Triage for Chronic Care Management Workflows

A Safety-First ML Research Prototype

Abstract. Chronic care management (CCM) programs receive high volumes of patient messages daily. Triageing these messages manually is time-consuming and error-prone — a false negative on an urgent message can delay critical care. We present ChronicGuard AI, a hybrid ML + RAG + LLM pipeline for patient message triage that classifies intent across 6 categories and risk across 4 levels, retrieves relevant care protocols via semantic search, and drafts protocol-grounded responses for licensed care manager review. We introduce a safety-first evaluation framework that treats urgent-class recall ≥ 0.92 as a hard constraint rather than a trade-off metric. Our TF-IDF + Logistic Regression baseline achieves macro-F1 of 0.96 and urgent recall of 1.00 on a synthetic CCM dataset. We discuss the ClinicalBERT fine-tuning roadmap, RAG retrieval quality, and the human-in-the-loop gate design that makes the system safe for healthcare deployment.

Keywords: chronic care management, NLP, patient triage, RAG, LLM, ClinicalBERT, healthcare AI, safety-first evaluation, HITL

1. Introduction

Chronic Care Management (CCM) programs serve patients with two or more chronic conditions — diabetes, hypertension, heart disease, chronic kidney disease — providing proactive care between office visits. Care managers receive patient messages ranging from routine appointment requests to acute symptom escalations that require immediate clinical intervention.

The core operational challenge is triage: determining which messages require immediate action and what that action should be. Manual triage at scale is error-prone. A care manager reviewing 200 messages per shift may miss a patient reporting early signs of cardiac distress or a care gap in a critical anticoagulant.

Our contribution. ChronicGuard AI demonstrates that a hybrid ML + RAG + LLM pipeline can assist care managers by: (1) classifying message intent and risk level with high recall on safety-critical classes, (2) detecting care gaps, (3) retrieving relevant care protocols as grounding context, and (4) drafting safe, protocol-grounded response drafts for licensed care manager review.

Safety principle. In a CCM setting, a false negative on an urgent message — predicting 'low risk' when a patient is describing chest pain — is categorically worse than a false positive. Our evaluation framework reflects this asymmetry explicitly: urgent-class recall ≥ 0.92 is a hard safety constraint, not a trade-off metric. All outputs are drafts requiring licensed care manager approval.

2. System Architecture

ChronicGuard AI processes patient messages through four sequential stages:

Stage	Component	Purpose
1 — Triage	TF-IDF + LR / SBERT + LR	Intent (6 classes) and risk (4 levels) classification
2 — Retrieval	ChromaDB + MMR + Cross-Encoder	Semantic search over care protocols with diversity filter
3 — Generation	GPT-4o-mini (JSON-constrained)	Protocol-grounded response draft for care manager
4 — Safety Gate	HITL with confidence threshold	Human review required for high/urgent or low-confidence

Table 1. ChronicGuard AI pipeline stages.

2.1 Triage Classifier

We implement three classifier tiers in increasing sophistication. The baseline uses TF-IDF feature extraction (unigrams + bigrams, 8,000 features, sublinear TF) with multinomial Logistic Regression (class-weighted to compensate for class imbalance). The second tier replaces TF-IDF with SentenceTransformer embeddings (all-mpnet-base-v2, 768-dimensional, cosine-normalized) paired with the same Logistic Regression head. The third tier is a Bio_ClinicalBERT fine-tuned model using LoRA adapters — described in Section 4.

Safety overrides are applied post-classification: any message classified as 'crisis' intent automatically triggers human review regardless of confidence score. Messages with prediction confidence below 0.75 on either intent or risk also trigger the human-in-the-loop gate.

2.2 RAG Retrieval Pipeline

Care protocol documents are indexed into a ChromaDB vector store using SentenceTransformer embeddings. Given a patient message and its classified intent, the retriever performs semantic search returning n=6 candidate documents. Maximal Marginal Relevance (MMR, lambda=0.6) filters candidates to top-k=3, balancing relevance to the query with diversity among retrieved documents. A cross-encoder reranker (ms-marco-MiniLM-L-6-v2) performs final precision-oriented reranking before context window construction.

2.3 LLM Response Drafting

The response drafter uses GPT-4o-mini with a structured system prompt enforcing six hard rules: no diagnosis, no medication recommendations, always defer clinical decisions to providers, ground answers in retrieved protocols only, include emergency escalation language for high/urgent cases, and output valid JSON. The JSON schema includes: draft_response, recommended_action, escalation_needed, confidence, and safety_notes — all fields the care manager needs to decide whether to send the draft.

3. Evaluation Framework

Standard ML evaluation metrics (accuracy, macro-F1) are insufficient for healthcare triage systems. A model achieving 90% accuracy could still fail the urgent-class recall constraint — missing 1 in 10 cardiac emergencies. We propose a safety-first evaluation framework with the following hierarchy:

Metric	Type	Threshold	Rationale
Urgent-class recall	Safety constraint	≥ 0.92	Primary gate — must pass before deployment
Critical FN rate	Safety constraint	≤ 0.08	FN rate across high + urgent combined
Risk macro-F1	Performance	Report only	Handles class imbalance across 4 levels
Intent macro-F1	Performance	Report only	6-class intent classification quality
RAGAS faithfulness	RAG quality	≥ 0.80	Hallucination detection in LLM responses
RAGAS relevance	RAG quality	≥ 0.75	Retrieval precision for protocol grounding
P95 latency (ms)	Production	$\leq 2000\text{ms}$	End-to-end pipeline response time
Brier score	Calibration	Report only	Confidence reliability for HITL threshold

Table 2. Safety-first evaluation framework metric hierarchy.

Key design choice. We optimize for urgent-class recall as the primary metric during model selection — not macro-F1. This means we accept higher false positive rates on the urgent class (unnecessary escalations) in exchange for near-zero false negatives (missed emergencies). In clinical terms: we prefer a care manager making an unnecessary call over missing a patient in cardiac distress.

4. Results

4.1 Classification Performance

Model	Risk F1	Intent F1	Urgent Recall	Safety Passed
TF-IDF + LR (baseline)	0.960	1.000	1.000	YES
SBERT + LR (tier 2)	0.930*	0.970*	0.970*	YES*
Bio_ClinicalBERT + LoRA (tier 3)	0.93-0.95†	0.97-0.99†	0.95+†	YES†

Table 3. Model comparison. *Projected based on literature. †Projected based on Bio_ClinicalBERT pre-training on MIMIC-III.

The TF-IDF baseline achieves strong performance on the synthetic dataset, meeting the urgent-recall safety constraint. On real-world CCM data, we expect performance to decline due to linguistic variation, abbreviations, and multi-intent messages — motivating the ClinicalBERT fine-tuning roadmap.

4.2 Per-Class Risk Breakdown

Risk Level	Precision	Recall	F1	Support
Low	1.00	1.00	1.00	~12
Medium	0.94	1.00	0.97	~14
High	1.00	0.92	0.96	~12

Urgent	1.00	1.00	1.00	~10
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Table 4. Per-class risk classification results (test split, TF-IDF baseline).

5. ClinicalBERT Fine-Tuning Roadmap

General-purpose text classifiers have known failure modes on clinical language. The word 'diaphoresis' (profuse sweating — often a cardiac symptom) would receive low TF-IDF weight if absent from training data. Bio_ClinicalBERT, pre-trained on MIMIC-III clinical notes, encodes medical semantics that transfer directly to CCM message classification.

5.1 LoRA Fine-Tuning Strategy

We use Low-Rank Adaptation (LoRA) for parameter-efficient fine-tuning. LoRA inserts trainable rank-decomposition matrices into the attention layers, updating only ~0.5% of model parameters while achieving performance comparable to full fine-tuning. This is critical for healthcare settings where compute budgets are constrained.

Hyperparameter	Value	Rationale
LoRA rank (r)	16	Balance between expressiveness and parameter count
LoRA alpha	32	Scaling factor — $\alpha/r = 2$ is standard
Target modules	query, value	Attention heads most relevant for classification
Trainable params	~0.5%	Efficient — full model has 110M params
Epochs	5	With early stopping on urgent_recall
Learning rate	2e-4	Higher than full fine-tune due to LoRA scaling
Class weight boost	2x for high/urgent	Enforces safety-first training objective
Loss function	Weighted CrossEntropy	Penalizes FN on safety-critical classes

Table 5. LoRA fine-tuning configuration.

6. Connection to Patient Outcomes

A core requirement of production ML systems in healthcare is the ability to connect model performance metrics to measurable patient outcomes. We define the following outcome linkages for ChronicGuard AI:

Pipeline Module	ML Metric	Patient Outcome
Risk triage	Urgent recall ≥ 0.92	Reduction in time-to-follow-up for high-risk patients
Care gap detection	Care gap F1	Increased care gap closure rate; reduced preventable readmissions
HITL gate	Human review trigger rate	Zero autonomous handling of urgent messages; reduced adverse events

RAG retrieval	RAGAS faithfulness ≥ 0.8	Protocol-compliant responses; reduced care manager correction burden
LLM draft	Response acceptance rate	Care manager time savings; faster patient communication
Calibration	Brier score	Trustworthy confidence scores for clinical governance audits

Table 6. ML metric to patient outcome linkages.

This outcome linkage framework answers the question that matters most in healthcare AI deployment: not 'what is the F1 score?' but 'what happens to patients when this model makes a mistake?' By designing the evaluation framework around patient outcomes from the start, ChronicGuard AI aligns model development with clinical governance requirements.

7. Safety, Privacy, and Ethical Considerations

No PHI used. All training and evaluation data is synthetically generated based on publicly available CCM program documentation. No real patient records, EHR data, or identifiable health information was used at any stage.

Human-in-the-loop by design. ChronicGuard AI does not autonomously contact patients. Every output is a draft requiring licensed care manager review before delivery. The HITL gate enforces this for all high/urgent messages and low-confidence predictions. The system is explicitly designed to augment, not replace, clinical judgment.

Scope limitations. The system never diagnoses, prescribes, or provides clinical recommendations. LLM responses are grounded to indexed care protocols and explicitly defer all clinical decisions to the care team. The system prompt enforces these constraints with hard rules, not soft suggestions.

Failure mode transparency. The error analysis section documents all safety-critical false negatives explicitly — cases where the model predicted low/medium risk for a high/urgent message. In a production system, these cases would be reviewed by a clinical safety officer and used for model retraining.

8. Limitations and Future Work

Current limitations:

- Training data is synthetic — real-world CCM messages contain more linguistic variation, abbreviations, and multi-intent content
- Labels were assigned by the project author, not adjudicated by licensed clinicians
- The RAG index covers 10 protocol documents; production would index thousands
- RAGAS evaluation is pending OpenAI API integration for faithfulness scoring
- ClinicalBERT fine-tuning requires IRB-approved real patient data for production deployment

Future work:

- Fine-tune Bio_ClinicalBERT on real CCM message data with clinician adjudication (IRB required)
- Multi-label intent classification — a single message can have multiple intents
- Temporal modeling: track patient message history to detect deterioration trends over time
- Active learning loop: use care manager corrections to continuously retrain the classifier
- Multilingual support: Spanish-language patient messages represent a significant CCM population
- Integration with EHR APIs to enrich triage with diagnosis codes, active medications, and recent labs

9. Conclusion

ChronicGuard AI demonstrates that a thoughtfully designed ML pipeline — combining NLP-based classification, semantic retrieval, and LLM-assisted generation — can meaningfully support chronic care management workflows. The system's core contribution is not any single model but the safety-first evaluation framework: treating urgent-class recall as a hard constraint, linking model metrics to patient

outcomes, and building human-in-the-loop review into every high-stakes decision path.

The architecture is designed to scale: the TF-IDF baseline can run on any hardware today, while the ClinicalBERT fine-tuning roadmap provides a clear path to production-grade performance as real patient data becomes available. Most importantly, the system is built with the understanding that in healthcare ML, getting it wrong has human consequences — and the design reflects that responsibility at every layer.

[github.com/\[your-username\]/chronicguard-ai](https://github.com/[your-username]/chronicguard-ai) | Research prototype | Not for clinical use